

VCS Task

I created a new local directory and added script files. Then I initialized it as a Git repository, created an empty GitHub repository, linked the local repository with GitHub using a remote origin, and pushed the code. I also added merge, rebase, and stash commands in the same repository.

Step-1:

Create a new directory and add script files with some content on it.

```
[sathish@Jeeva-M1 GuviTraining % mkdir vsctask
[sathish@Jeeva-M1 GuviTraining % cd vsctask
[sathish@Jeeva-M1 vsctask % touch script1.txt script2.txt script3.txt
[sathish@Jeeva-M1 vsctask % echo "Hello from script1" > script1.txt
[sathish@Jeeva-M1 vsctask % echo "Hello from script2" > script12.txt
[sathish@Jeeva-M1 vsctask % echo "Hello from script3" > script3.txt
[sathish@Jeeva-M1 vsctask % ls
script1.txt      script12.txt      script2.txt      script3.txt
[sathish@Jeeva-M1 vsctask % rm script12.txt
[sathish@Jeeva-M1 vsctask % ls
script1.txt      script2.txt      script3.txt
```

Initialize local Git repository

```
[sathish@Jeeva-M1 vsctask % git init
hint: Using 'master' as the name for the initial branch. This default branch name
hint: will change to "main" in Git 3.0. To configure the initial branch name
hint: to use in all of your new repositories, which will suppress this warning,
hint: call:
hint:   git config --global init.defaultBranch <name>
hint: Names commonly chosen instead of 'master' are 'main', 'trunk' and
hint: 'development'. The just-created branch can be renamed via this command:
hint:   git branch -m <name>
hint: Disable this message with "git config set advice.defaultBranchName false"
Initialized empty Git repository in /Users/sathish/Documents/GuviTraining/vsctask/.git/
[sathish@Jeeva-M1 vsctask % git add .
```

Add and commit the files.

```
[sathish@Jeeva-M1 vsctask % git status]
On branch master

No commits yet

Changes to be committed:
  (use "git rm --cached <file>..." to unstage)
        new file:   script1.txt
        new file:   script2.txt
        new file:   script3.txt

[sathish@Jeeva-M1 vsctask % git commit -m "vcs commit"]
[master (root-commit) 1374ab3] vcs commit
 3 files changed, 3 insertions(+)
 create mode 100644 script1.txt
 create mode 100644 script2.txt
 create mode 100644 script3.txt
sathish@Jeeva-M1 vsctask %
```

Step-2:

Create empty GitHub repository and link local repo to GitHub

```
[sathish@Jeeva-M1 vsctask % git remote add origin https://github.com/SathishNJ/vcs_tasks.git]
[sathish@Jeeva-M1 vsctask % git remote -v]
origin https://github.com/SathishNJ/vcs_tasks.git (fetch)
origin https://github.com/SathishNJ/vcs_tasks.git (push)
[sathish@Jeeva-M1 vsctask % git branch]
* master
```

Push code to GitHub

```
[sathish@Jeeva-M1 vsctask % git push -u origin master]
Username for 'https://github.com': SathishNJ
[Password for 'https://SathishNJ@github.com':]
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Delta compression using up to 8 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (5/5), 343 bytes | 343.00 KiB/s, done.
Total 5 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/SathishNJ/vcs_tasks.git
 * [new branch]      master -> master
branch 'master' set up to track 'origin/master'.
sathish@Jeeva-M1 vsctask %
```

Step - 3:

Create a new branch to work on merge commands.

Checkout out the new branch, modify the script file, and commit the changes. Switch to the master branch and merge the new branch into the master branch.

```

[sathish@Jeeva-M1 vsctask % git checkout -b feature-branch
Switched to a new branch 'feature-branch'
[sathish@Jeeva-M1 vsctask % echo "Example for merging" > script1.txt
[sathish@Jeeva-M1 vsctask % git add .
[sathish@Jeeva-M1 vsctask % git branch
* feature-branch
master
[sathish@Jeeva-M1 vsctask % git commit -m "Modified script1 for merge"
[feature-branch 2e19ddb] Modified script1 for merge
1 file changed, 1 insertion(+), 1 deletion(-)
[sathish@Jeeva-M1 vsctask % git checkout master
Switched to branch 'master'
Your branch is up to date with 'origin/master'.
[sathish@Jeeva-M1 vsctask % git merge feature-branch
Updating 1374ab3..2e19ddb
Fast-forward
 script1.txt | 2 +-
1 file changed, 1 insertion(+), 1 deletion(-)

```

Push the changes to the master branch.

```

[sathish@Jeeva-M1 vsctask % git push origin master
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Delta compression using up to 8 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (3/3), 332 bytes | 332.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/SathishNJ/vcs_tasks.git
1374ab3..2e19ddb master -> master
[sathish@Jeeva-M1 vsctask %

```

Step - 4:

Create a new branch to work on rebase commands.

Checkout out the new branch, modify the script file, and commit the changes.

```

[sathish@Jeeva-M1 vsctask % git checkout -b feature-rebase
Switched to a new branch 'feature-rebase'
[sathish@Jeeva-M1 vsctask % echo "Rebase script change" > script2.txt
[sathish@Jeeva-M1 vsctask % git add .
[sathish@Jeeva-M1 vsctask % git commit -m "rebase example"
[feature-rebase d04814d] rebase example
1 file changed, 1 insertion(+), 1 deletion(-)

```

Switch to the master branch, modify the script file, and commit the changes

```
[sathish@Jeeva-M1 vsctask % git checkout master
Switched to branch 'master'
Your branch is up to date with 'origin/master'.
[sathish@Jeeva-M1 vsctask % echo "update master branch" > script1.txt
[sathish@Jeeva-M1 vsctask % git add .
[sathish@Jeeva-M1 vsctask % git commit -m "Updated script in master"
[master 21c8cb4] Updated script in master
1 file changed, 1 insertion(+), 1 deletion(-)
```

Now switch to feature-rebase branch and rebase with master branch, and push the branch.

```
[sathish@Jeeva-M1 vsctask % git checkout feature-rebase
Switched to branch 'feature-rebase'
[sathish@Jeeva-M1 vsctask % git status
On branch feature-rebase
nothing to commit, working tree clean
[sathish@Jeeva-M1 vsctask % git rebase master
Successfully rebased and updated refs/heads/feature-rebase.
[sathish@Jeeva-M1 vsctask % git push origin feature-rebase
Enumerating objects: 9, done.
Counting objects: 100% (9/9), done.
Delta compression using up to 8 threads
Compressing objects: 100% (4/4), done.
Writing objects: 100% (6/6), 629 bytes | 629.00 KiB/s, done.
Total 6 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
remote:
remote: Create a pull request for 'feature-rebase' on GitHub by visiting:
remote:      https://github.com/SathishNJ/vcs_tasks/pull/new/feature-rebase
remote:
To https://github.com/SathishNJ/vcs_tasks.git
* [new branch]      feature-rebase -> feature-rebase
```

Step - 5:

Create a new branch to work on stash commands.

Checkout out the new branch.

```
[sathish@Jeeva-M1 vsctask % git checkout master
Switched to branch 'master'
Your branch is ahead of 'origin/master' by 1 commit.
(use "git push" to publish your local commits)
[sathish@Jeeva-M1 vsctask % git checkout -m feature-stash
error: pathspec 'feature-stash' did not match any file(s) known to git
[sathish@Jeeva-M1 vsctask % git checkout -b feature-stash
Switched to a new branch 'feature-stash'
```

Make some changes to the script file, but do not commit.

```
[sathish@Jeeva-M1 vsctask % echo "---Stash example ---" > script1.txt
[sathish@Jeeva-M1 vsctask % git status
On branch feature-stash
Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
        modified:   script1.txt

no changes added to commit (use "git add" and/or "git commit -a")
sathish@Jeeva-M1 vsctask % █
```

Stash the changes

```
[sathish@Jeeva-M1 vsctask % git stash
Saved working directory and index state WIP on feature-stash: 21c8cb4 Updated sc
ript in master
[sathish@Jeeva-M1 vsctask % git stash list
stash@{0}: WIP on feature-stash: 21c8cb4 Updated script in master
sathish@Jeeva-M1 vsctask % █
```

Apply the stash

```
[sathish@Jeeva-M1 vsctask % git stash apply
On branch feature-stash
Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
        modified:   script1.txt

no changes added to commit (use "git add" and/or "git commit -a")
```

Commit the restored changes, and push the changes to the master branch.

```
[sathish@Jeeva-M1 vsctask % git add .
[sathish@Jeeva-M1 vsctask % git commit -m "Apply stashed changes"
[feature-stash 6512c56] Apply stashed changes
 1 file changed, 1 insertion(+), 1 deletion(-)
[sathish@Jeeva-M1 vsctask % git push origin master
Total 0 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/SathishNJ/vcs_tasks.git
   2e19ddb..21c8cb4  master -> master
[sathish@Jeeva-M1 vsctask % git branch
  feature-branch
  feature-rebase
* feature-stash
  master
sathish@Jeeva-M1 vsctask % █
```